

Prelim-Code-P3.R

stein

2026-04-21

```
library(survminer)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: ggpubr
```

```
library(survival)
```

```
##
```

```
## Attaching package: 'survival'
```

```
## The following object is masked from 'package:survminer':
```

```
##
```

```
##      myeloma
```

```
library(gtsummary)
```

```
library(MASS)
```

```
##
```

```
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:gtsummary':
```

```
##
```

```
##      select
```

```
library(gt)
```

```
library(knitr)
```

```
# Upload Data set
```

```
fhsdata <- read.csv("C:/Users/stein/OneDrive/Documents/School/2026 Spring/Advanced Methods/BIOS6624/Pro
```

```
# Notes to self:
```

```
# remove stroke = 1 at baseline
```

```
# remove stroke >3652.5 days
```

```
# check missing: BP(systolic), age, diabetes
```

```

# heart disease, smoking status, bp medication, HDL/LDL, BMI
#

## Initial data cleaning

# remove anyone with stroke at baseline (STROKE = 1 at period 1)
clean <- fhsdata[!(fhsdata$TIMESTRK == 0 & fhsdata$PERIOD == 1), ]

# filter to use just the baseline values
base <- clean[clean$PERIOD == 1, ]

# censor follow-up to 10 years (365.25*10)
base$stroke_event <- ifelse(base$TIMESTRK <= 3652.5 & base$STROKE == 1, 1, 0)
base$stroke_time <- ifelse(base$TIMESTRK <= 3652.5, base$TIMESTRK, 3652.5)

# check covariates for missing
sum(is.na(base$SYSBP)) # 0

```

```
## [1] 0
```

```
sum(is.na(base$AGE)) # 0
```

```
## [1] 0
```

```
sum(is.na(base$DIABETES)) # 0
```

```
## [1] 0
```

```
sum(is.na(base$CURSMOKE)) # 0
```

```
## [1] 0
```

```
sum(is.na(base$BMI)) # 17
```

```
## [1] 17
```

```
sum(is.na(base$TOTCHOL)) # 52
```

```
## [1] 52
```

```
sum(is.na(base$BPMEDS)) # 60
```

```
## [1] 60
```

```
sum(is.na(base$HEARTRTE)) # 0
```

```
## [1] 1
```

```

# check covariates for implausible values
sum(base$SYSBP > 300 | base$SYSBP < 60) # 0

## [1] 0

sum(base$AGE > 99 | base$AGE <= 18) # 0

## [1] 0

min(base$BMI, na.rm = TRUE) # 15.54

## [1] 15.54

max(base$BMI, na.rm = TRUE) # 56.8 all plausible values for BMI

## [1] 56.8

#####
## Table 1
#####
library(gtsummary)

table1_data <- base

table1_data$SEX      <- factor(base$SEX,      levels = c(1, 2), labels = c("Male", "Female"))
table1_data$DIABETES <- factor(base$DIABETES, levels = c(0, 1), labels = c("No", "Yes"))
table1_data$CURSMOKE <- factor(base$CURSMOKE, levels = c(0, 1), labels = c("No", "Yes"))
table1_data$stroke_event <- factor(base$stroke_event, levels = c(0, 1), labels = c("No", "Yes"))

# Build Table 1 - only variables in final model + age + outcome
table1 <- tbl_summary(
  data = table1_data[, c("SEX", "AGE", "SYSBP", "DIABETES", "CURSMOKE", "stroke_event")],
  by = SEX,
  label = list(
    AGE      ~ "Age (years)",
    SYSBP     ~ "Systolic Blood Pressure (mmHg)",
    DIABETES  ~ "Diabetes",
    CURSMOKE  ~ "Current Smoker",
    stroke_event ~ "Stroke Event within 10 Years"
  ),
  statistic = list(
    all_continuous() ~ "{mean} ({sd})",
    all_categorical() ~ "{n} ({p}%)"
  ),
  digits = list(
    all_continuous() ~ 1,
    all_categorical() ~ c(0, 1)
  ),
  missing = "ifany",
  missing_text = "Missing"
) %>%

```

```

add_overall() %>%
bold_labels() %>%
modify_header(label ~ "**Characteristic**") %>%
modify_caption("**Table 1. Baseline Characteristics by Sex**")

print(table1)

```

```

## <div id="pmyztlapan" style="padding-left:0px;padding-right:0px;padding-top:10px;padding-bottom:10px;
##   <style>#pmyztlapan table {
##     font-family: system-ui, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif, 'Apple Color Emoji', 'Se
##     -webkit-font-smoothing: antialiased;
##     -moz-osx-font-smoothing: grayscale;
##   }
##
## #pmyztlapan thead, #pmyztlapan tbody, #pmyztlapan tfoot, #pmyztlapan tr, #pmyztlapan td, #pmyztlapan
##   border-style: none;
## }
##
## #pmyztlapan p {
##   margin: 0;
##   padding: 0;
## }
##
## #pmyztlapan .gt_table {
##   display: table;
##   border-collapse: collapse;
##   line-height: normal;
##   margin-left: auto;
##   margin-right: auto;
##   color: #333333;
##   font-size: 16px;
##   font-weight: normal;
##   font-style: normal;
##   background-color: #FFFFFF;
##   width: auto;
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #A8A8A8;
##   border-right-style: none;
##   border-right-width: 2px;
##   border-right-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #A8A8A8;
##   border-left-style: none;
##   border-left-width: 2px;
##   border-left-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_caption {
##   padding-top: 4px;
##   padding-bottom: 4px;

```

```

## }
##
## #pmyztlapan .gt_title {
##   color: #333333;
##   font-size: 125%;
##   font-weight: initial;
##   padding-top: 4px;
##   padding-bottom: 4px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-bottom-color: #FFFFFF;
##   border-bottom-width: 0;
## }
##
## #pmyztlapan .gt_subtitle {
##   color: #333333;
##   font-size: 85%;
##   font-weight: initial;
##   padding-top: 3px;
##   padding-bottom: 5px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-top-color: #FFFFFF;
##   border-top-width: 0;
## }
##
## #pmyztlapan .gt_heading {
##   background-color: #FFFFFF;
##   text-align: center;
##   border-bottom-color: #FFFFFF;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;
##   border-right-style: none;
##   border-right-width: 1px;
##   border-right-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_bottom_border {
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_col_headings {
##   border-top-style: solid;
##   border-top-width: 2px;
##   border-top-color: #D3D3D3;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
##   border-left-style: none;
##   border-left-width: 1px;
##   border-left-color: #D3D3D3;

```

```

## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_col_heading {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 6px;
## padding-left: 5px;
## padding-right: 5px;
## overflow-x: hidden;
## }
##
## #pmyztlapan .gt_column_spanner_outer {
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: normal;
## text-transform: inherit;
## padding-top: 0;
## padding-bottom: 0;
## padding-left: 4px;
## padding-right: 4px;
## }
##
## #pmyztlapan .gt_column_spanner_outer:first-child {
## padding-left: 0;
## }
##
## #pmyztlapan .gt_column_spanner_outer:last-child {
## padding-right: 0;
## }
##
## #pmyztlapan .gt_column_spanner {
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## vertical-align: bottom;
## padding-top: 5px;
## padding-bottom: 5px;
## overflow-x: hidden;
## display: inline-block;

```

```

## width: 100%;
## }
##
## #pmyztlapan .gt_spanner_row {
## border-bottom-style: hidden;
## }
##
## #pmyztlapan .gt_group_heading {
## padding-top: 8px;
## padding-bottom: 8px;
## padding-left: 5px;
## padding-right: 5px;
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: initial;
## text-transform: inherit;
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 1px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 1px;
## border-right-color: #D3D3D3;
## vertical-align: middle;
## text-align: left;
## }
##
## #pmyztlapan .gt_empty_group_heading {
## padding: 0.5px;
## color: #333333;
## background-color: #FFFFFF;
## font-size: 100%;
## font-weight: initial;
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## vertical-align: middle;
## }
##
## #pmyztlapan .gt_from_md > :first-child {
## margin-top: 0;
## }
##
## #pmyztlapan .gt_from_md > :last-child {
## margin-bottom: 0;

```

```

## }
##
## #pmyztlapan .gt_row {
##     padding-top: 8px;
##     padding-bottom: 8px;
##     padding-left: 5px;
##     padding-right: 5px;
##     margin: 10px;
##     border-top-style: solid;
##     border-top-width: 1px;
##     border-top-color: #D3D3D3;
##     border-left-style: none;
##     border-left-width: 1px;
##     border-left-color: #D3D3D3;
##     border-right-style: none;
##     border-right-width: 1px;
##     border-right-color: #D3D3D3;
##     vertical-align: middle;
##     overflow-x: hidden;
## }
##
## #pmyztlapan .gt_stub {
##     color: #333333;
##     background-color: #FFFFFF;
##     font-size: 100%;
##     font-weight: initial;
##     text-transform: inherit;
##     border-right-style: solid;
##     border-right-width: 2px;
##     border-right-color: #D3D3D3;
##     padding-left: 5px;
##     padding-right: 5px;
## }
##
## #pmyztlapan .gt_stub_row_group {
##     color: #333333;
##     background-color: #FFFFFF;
##     font-size: 100%;
##     font-weight: initial;
##     text-transform: inherit;
##     border-right-style: solid;
##     border-right-width: 2px;
##     border-right-color: #D3D3D3;
##     padding-left: 5px;
##     padding-right: 5px;
##     vertical-align: top;
## }
##
## #pmyztlapan .gt_row_group_first td {
##     border-top-width: 2px;
## }
##
## #pmyztlapan .gt_row_group_first th {
##     border-top-width: 2px;

```

```

## }
##
## #pmyztlapan .gt_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #pmyztlapan .gt_first_summary_row {
##   border-top-style: solid;
##   border-top-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_first_summary_row.thick {
##   border-top-width: 2px;
## }
##
## #pmyztlapan .gt_last_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-bottom-style: solid;
##   border-bottom-width: 2px;
##   border-bottom-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_grand_summary_row {
##   color: #333333;
##   background-color: #FFFFFF;
##   text-transform: inherit;
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #pmyztlapan .gt_first_grand_summary_row {
##   padding-top: 8px;
##   padding-bottom: 8px;
##   padding-left: 5px;
##   padding-right: 5px;
##   border-top-style: double;
##   border-top-width: 6px;
##   border-top-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_last_grand_summary_row_top {
##   padding-top: 8px;
##   padding-bottom: 8px;

```

```

## padding-left: 5px;
## padding-right: 5px;
## border-bottom-style: double;
## border-bottom-width: 6px;
## border-bottom-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_stripped {
## background-color: rgba(128, 128, 128, 0.05);
## }
##
## #pmyztlapan .gt_table_body {
## border-top-style: solid;
## border-top-width: 2px;
## border-top-color: #D3D3D3;
## border-bottom-style: solid;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_footnotes {
## color: #333333;
## background-color: #FFFFFF;
## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;
## border-right-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_footnote {
## margin: 0px;
## font-size: 90%;
## padding-top: 4px;
## padding-bottom: 4px;
## padding-left: 5px;
## padding-right: 5px;
## }
##
## #pmyztlapan .gt_sourcenotes {
## color: #333333;
## background-color: #FFFFFF;
## border-bottom-style: none;
## border-bottom-width: 2px;
## border-bottom-color: #D3D3D3;
## border-left-style: none;
## border-left-width: 2px;
## border-left-color: #D3D3D3;
## border-right-style: none;
## border-right-width: 2px;

```

```

##   border-right-color: #D3D3D3;
## }
##
## #pmyztlapan .gt_sourcenote {
##   font-size: 90%;
##   padding-top: 4px;
##   padding-bottom: 4px;
##   padding-left: 5px;
##   padding-right: 5px;
## }
##
## #pmyztlapan .gt_left {
##   text-align: left;
## }
##
## #pmyztlapan .gt_center {
##   text-align: center;
## }
##
## #pmyztlapan .gt_right {
##   text-align: right;
##   font-variant-numeric: tabular-nums;
## }
##
## #pmyztlapan .gt_font_normal {
##   font-weight: normal;
## }
##
## #pmyztlapan .gt_font_bold {
##   font-weight: bold;
## }
##
## #pmyztlapan .gt_font_italic {
##   font-style: italic;
## }
##
## #pmyztlapan .gt_super {
##   font-size: 65%;
## }
##
## #pmyztlapan .gt_footnote_marks {
##   font-size: 75%;
##   vertical-align: 0.4em;
##   position: initial;
## }
##
## #pmyztlapan .gt_asterisk {
##   font-size: 100%;
##   vertical-align: 0;
## }
##
## #pmyztlapan .gt_indent_1 {
##   text-indent: 5px;
## }

```

```

##
## #pmyztlapan .gt_indent_2 {
##   text-indent: 10px;
## }
##
## #pmyztlapan .gt_indent_3 {
##   text-indent: 15px;
## }
##
## #pmyztlapan .gt_indent_4 {
##   text-indent: 20px;
## }
##
## #pmyztlapan .gt_indent_5 {
##   text-indent: 25px;
## }
##
## #pmyztlapan .katex-display {
##   display: inline-flex !important;
##   margin-bottom: 0.75em !important;
## }
##
## #pmyztlapan div.Reactable > div.rt-table > div.rt-thead > div.rt-tr.rt-tr-group-header > div.rt-th-g
##   height: 0px !important;
## }
## </style>
## <table class="gt_table" data-quarto-disable-processing="false" data-quarto-bootstrap="false">
##   <caption><span class='gt_from_md'><strong>Table 1. Baseline Characteristics by Sex</strong></span>
##   <thead>
##     <tr class="gt_col_headings">
##       <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col
##       <th class="gt_col_heading gt_columns_bottom_border gt_center" rowspan="1" colspan="1" scope="c
##       <th class="gt_col_heading gt_columns_bottom_border gt_center" rowspan="1" colspan="1" scope="c
##       <th class="gt_col_heading gt_columns_bottom_border gt_center" rowspan="1" colspan="1" scope="c
##     </tr>
##   </thead>
##   <tbody class="gt_table_body">
##     <tr><td headers="label" class="gt_row gt_left" style="font-weight: bold;">Age (years)</td>
## <td headers="stat_0" class="gt_row gt_center">49.9 (8.7)</td>
## <td headers="stat_1" class="gt_row gt_center">49.7 (8.7)</td>
## <td headers="stat_2" class="gt_row gt_center">50.0 (8.6)</td></tr>
##     <tr><td headers="label" class="gt_row gt_left" style="font-weight: bold;">Systolic Blood Pressur
## <td headers="stat_0" class="gt_row gt_center">132.8 (22.3)</td>
## <td headers="stat_1" class="gt_row gt_center">131.6 (19.4)</td>
## <td headers="stat_2" class="gt_row gt_center">133.7 (24.3)</td></tr>
##     <tr><td headers="label" class="gt_row gt_left" style="font-weight: bold;">Diabetes</td>
## <td headers="stat_0" class="gt_row gt_center">119 (2.7%)</td>
## <td headers="stat_1" class="gt_row gt_center">59 (3.1%)</td>
## <td headers="stat_2" class="gt_row gt_center">60 (2.4%)</td></tr>
##     <tr><td headers="label" class="gt_row gt_left" style="font-weight: bold;">Current Smoker</td>
## <td headers="stat_0" class="gt_row gt_center">2,169 (49.3%)</td>
## <td headers="stat_1" class="gt_row gt_center">1,168 (60.5%)</td>
## <td headers="stat_2" class="gt_row gt_center">1,001 (40.5%)</td></tr>
##     <tr><td headers="label" class="gt_row gt_left" style="font-weight: bold;">Stroke Event within 10

```

Table 1: Table 1. Baseline Characteristics by Sex

Characteristic	Overall N = 4,402 ¹	Male N = 1,930 ¹	Female N = 2,472 ¹
Age (years)	49.9 (8.7)	49.7 (8.7)	50.0 (8.6)
Systolic Blood Pressure (mmHg)	132.8 (22.3)	131.6 (19.4)	133.7 (24.3)
Diabetes	119 (2.7%)	59 (3.1%)	60 (2.4%)
Current Smoker	2,169 (49.3%)	1,168 (60.5%)	1,001 (40.5%)
Stroke Event within 10 Years	111 (2.5%)	49 (2.5%)	62 (2.5%)

¹Mean (SD); n (%)

```
## <td headers="stat_0" class="gt_row gt_center">111 (2.5%)</td>
## <td headers="stat_1" class="gt_row gt_center">49 (2.5%)</td>
## <td headers="stat_2" class="gt_row gt_center">62 (2.5%)</td></tr>
## </tbody>
## <tfoot>
##   <tr class="gt_footnotes">
##     <td class="gt_footnote" colspan="4"><span class="gt_footnote_marks" style="white-space:nowrap;
##   </tr>
## </tfoot>
## </table>
## </div>
```

table1

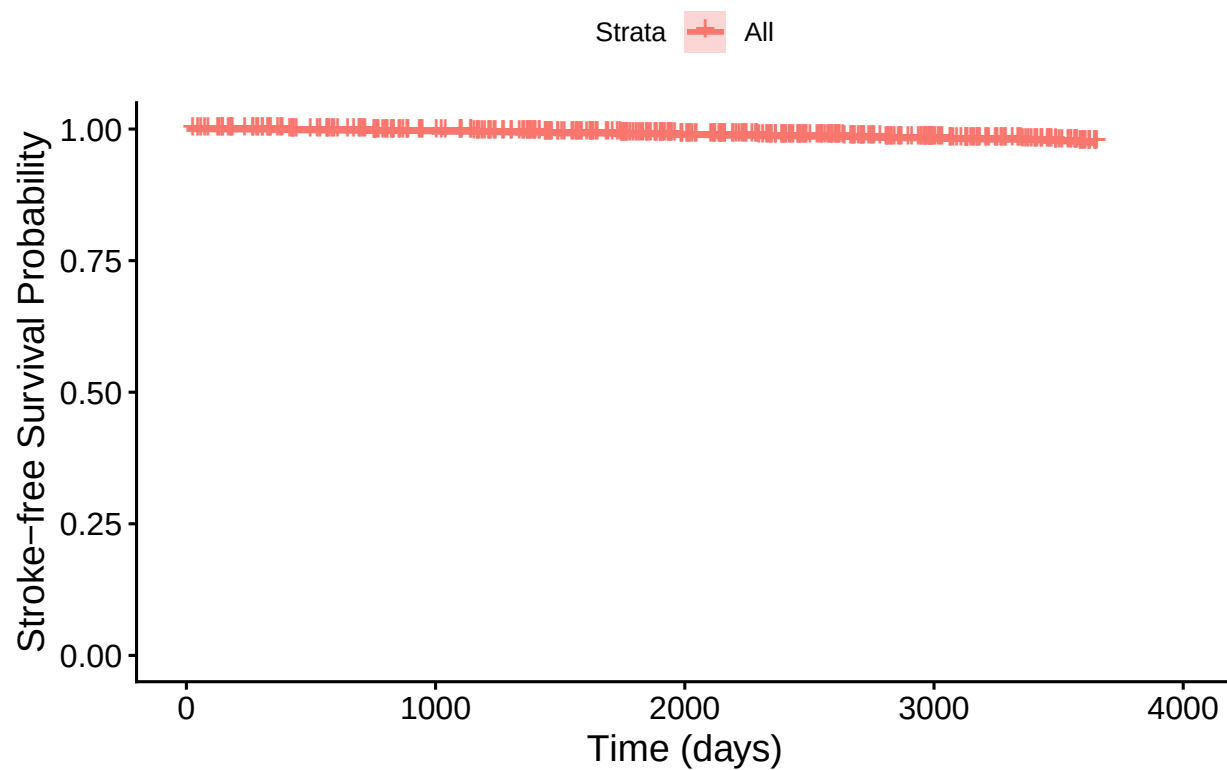
```
### fit KM curves to determine what we put in the cox model

surv_obj <- Surv(time = base$stroke_time, event = base$stroke_event)

# not stratified
km_overall <- survfit(surv_obj ~ 1, data = base)

ggsurvplot(km_overall,
            data = base,
            xlab = "Time (days)",
            ylab = "Stroke-free Survival Probability",
            title = "Overall Kaplan-Meier Curve")
```

Overall Kaplan–Meier Curve

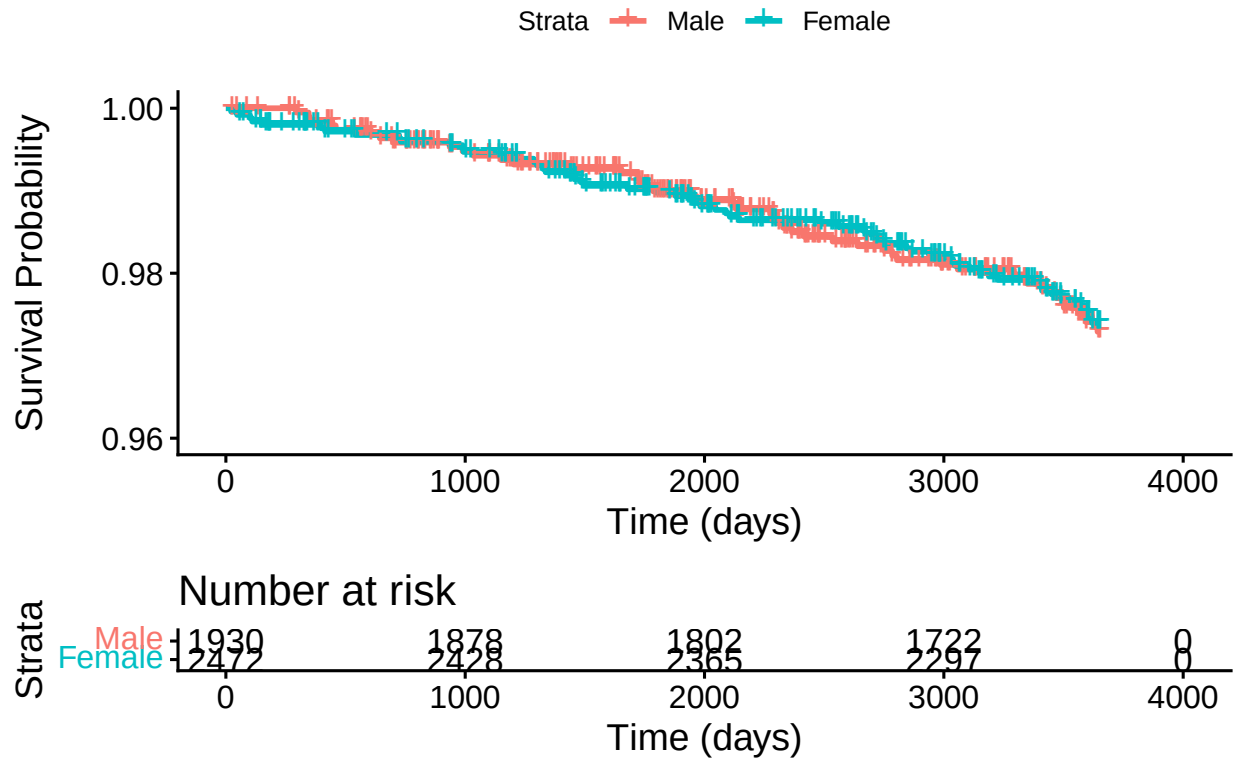


```
# stratified by sex
km_sex <- survfit(surv_obj ~ SEX, data = base)

ggsurvplot(km_sex,
  data = base,
  legend.labs = c("Male", "Female"),
  xlab = "Time (days)",
  ylab = "Survival Probability",
  title = "Kaplan-Meier by Sex",
  ylim = c(0.96, 1),
  break.y.by = 0.02,
  risk.table = T)
```

```
## Ignoring unknown labels:
## * colour : "Strata"
```

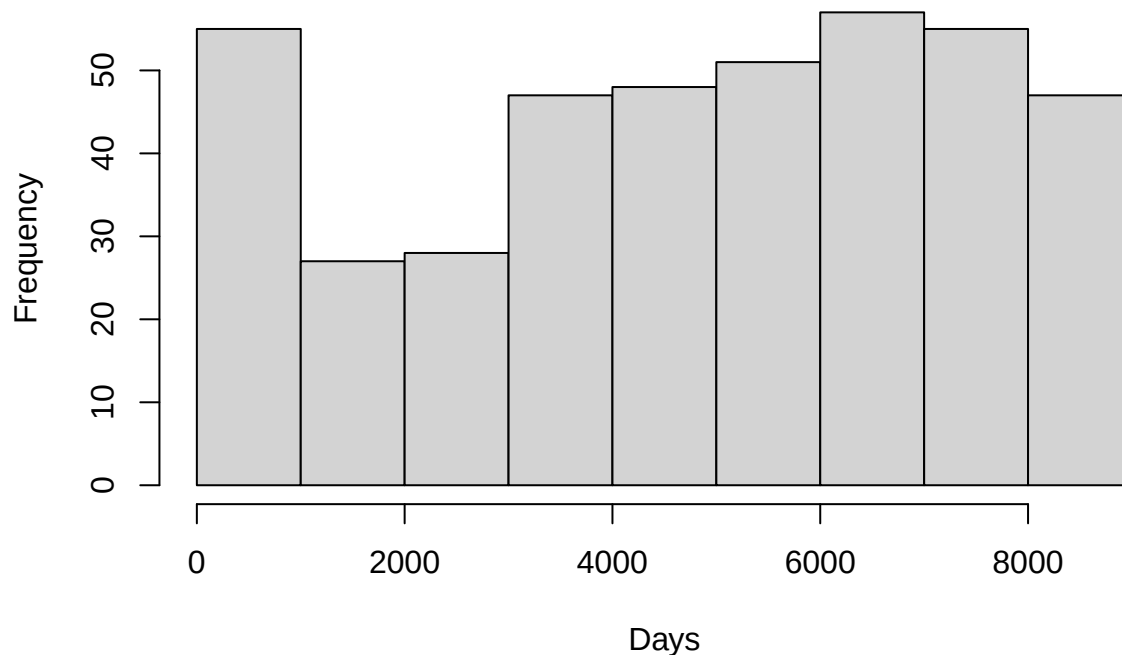
Kaplan–Meier by Sex



FILTERED DATA WRONG ##### fix it

```
hist(fhsdata$TIMESTRK[fhsdata$PERIOD == 1 & fhsdata$STROKE == 1],
     main = "TIMESTRK for Stroke=1 at Period 1",
     xlab = "Days")
```

TIMESTRK for Stroke=1 at Period 1



```
summary(fhsdata$TIMESTRK[fhsdata$PERIOD == 1 & fhsdata$STROKE == 1])
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##         0   2769    5134   4740   6952   8702
```

```
### check the variables that we are debating including
```

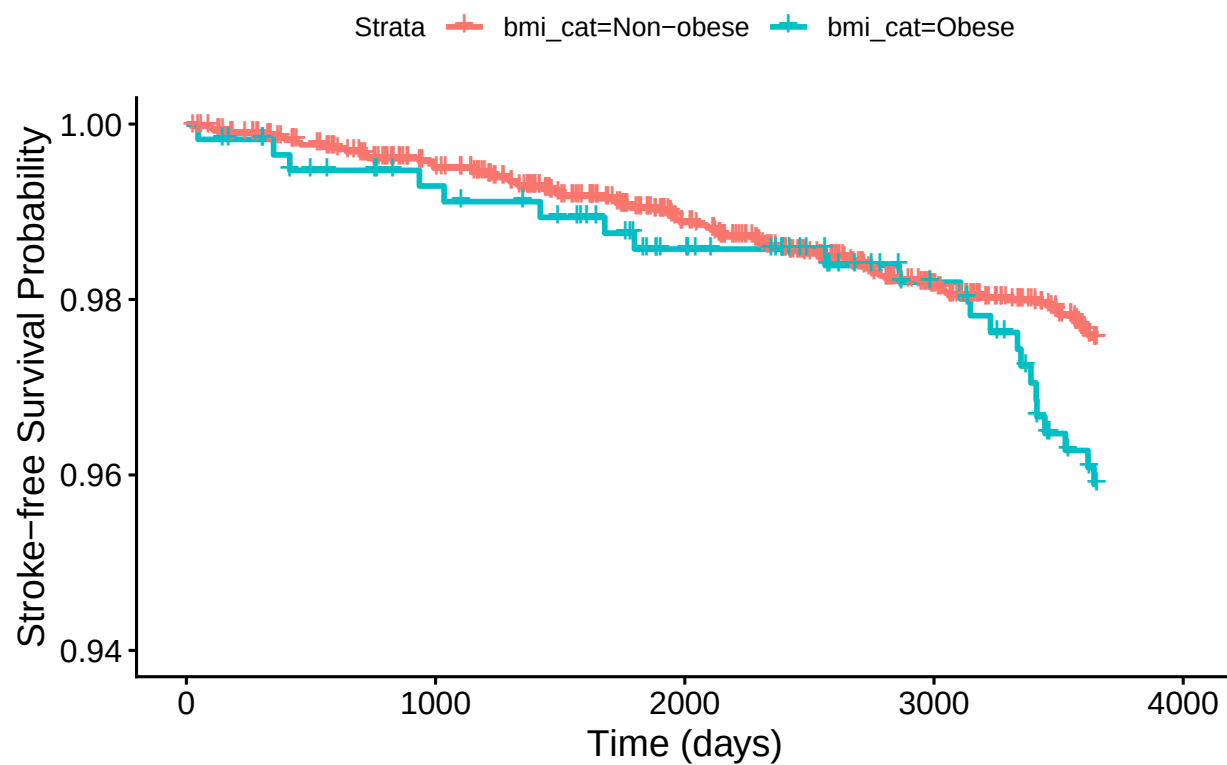
```
# Dichotomized versions are just for KM plotting
```

```
base$bmi_cat    <- ifelse(base$BMI >= 30, "Obese", "Non-obese")
base$chol_cat   <- ifelse(base$TOTCHOL >= 240, "High Chol", "Normal Chol")
base$bpmeds_cat <- ifelse(base$BPMEDS == 1, "On BP Meds", "Not on BP Meds")
base$hearttrte_cat <- ifelse(base$HEARTRTE > 100, "High HR", "Normal HR")
base$smoke_cat  <- ifelse(base$CURSMOKE == 1, "Smoker", "Non-smoker")
```

```
# BMI
```

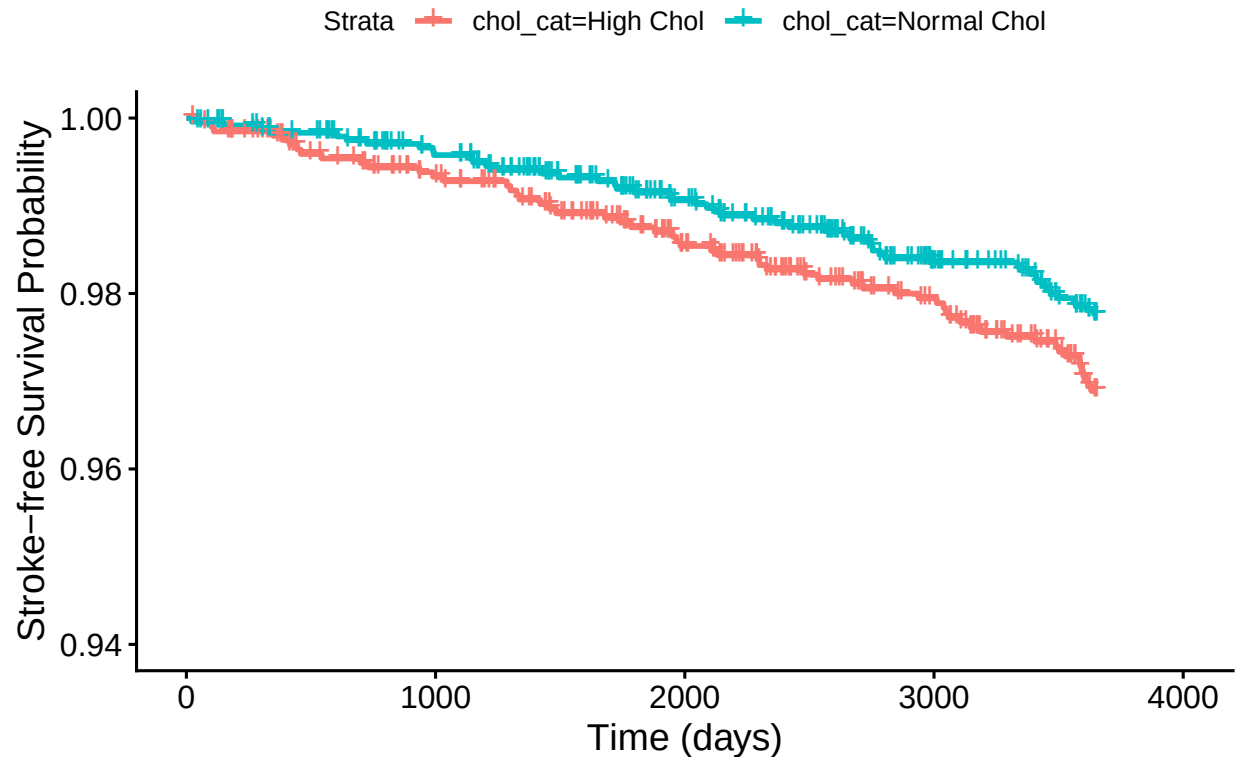
```
km_bmi <- survfit(surv_obj ~ bmi_cat, data = base)
ggsurvplot(km_bmi, data = base,,
  ylim = c(0.94, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by BMI")
```

KM Curve by BMI



```
# Total cholesterol
km_chol <- survfit(surv_obj ~ chol_cat, data = base)
ggsurvplot(km_chol, data = base,
  ylim = c(0.94, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by Total Cholesterol")
```

KM Curve by Total Cholesterol



```
# BP meds
km_bpmeds <- survfit(surv_obj ~ bpmeds_cat, data = base)
ggsurvplot(km_bpmeds, data = base,
  ylim = c(0.94, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by BP Meds")
```

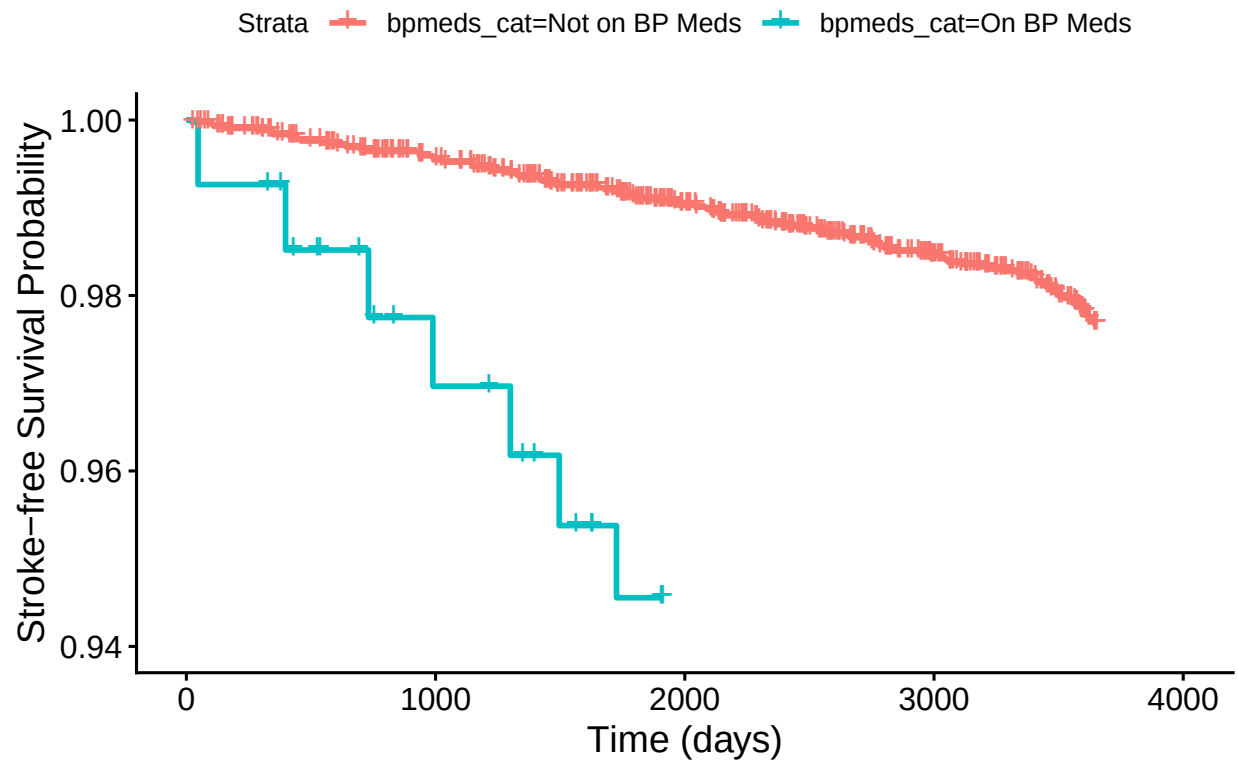
```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_step()').
```

```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_step()').
```

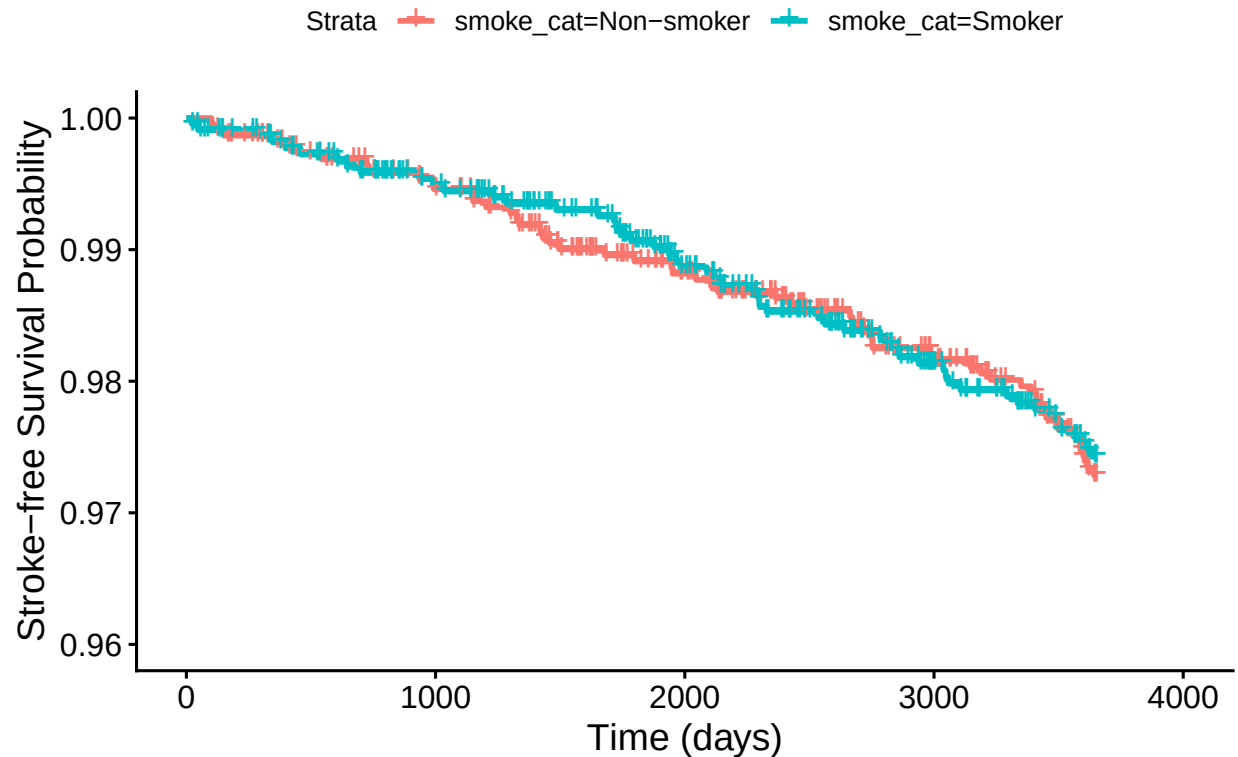
```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_point()').
```

KM Curve by BP Meds



```
# Smoking
km_smoke <- survfit(surv_obj ~ smoke_cat, data = base)
ggsurvplot(km_smoke, data = base,
  ylim = c(0.96, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by Smoking Status")
```

KM Curve by Smoking Status



```
# Heart rate
km_hearttrte <- survfit(surv_obj ~ hearttrte_cat, data = base)
ggsurvplot(km_hearttrte, data = base,
  ylim = c(0.96, 1),
  xlab = "Time (days)",
  ylab = "Stroke-free Survival Probability",
  title = "KM Curve by Heart Rate")
```

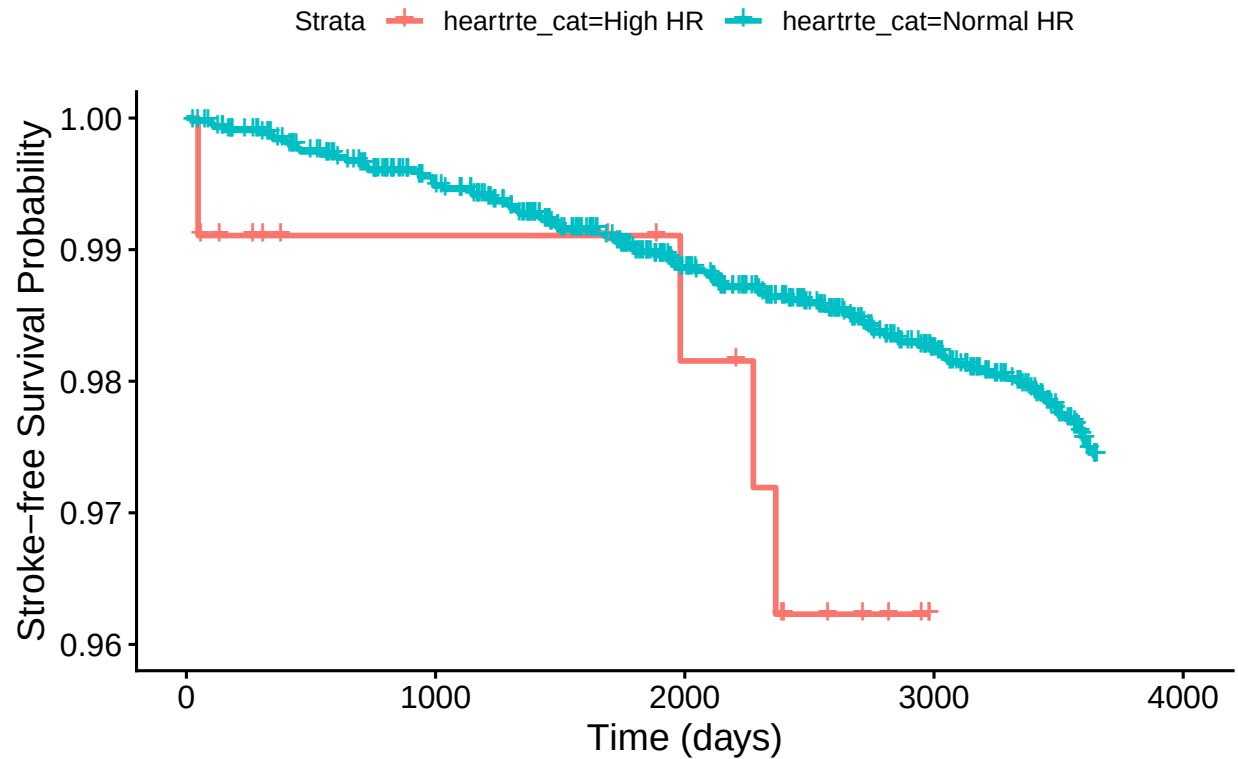
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_step()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_step()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```

KM Curve by Heart Rate



```
### Start survival models ###
```

```
### complete case analysis to add back in dropped variables
```

```
base_final <- base[complete.cases(base[, c("SYSBP", "AGE",
                                             "DIABETES", "BPMEDS"))], ]
nrow(base_final)
```

```
## [1] 4342
```

```
sapply(base[, c("SYSBP", "AGE",
                 "DIABETES", "BPMEDS")], function(x) sum(is.na(x)))
```

```
##      SYSBP      AGE DIABETES      BPMEDS
##         0         0         0         60
```

```
## split into male and female
```

```
base_male <- base_final[base_final$SEX == 1, ]
base_female <- base_final[base_final$SEX == 2, ]
```

```
## update survival objects for both
```

```

surv_male <- Surv(time = base_male$stroke_time,
                  event = base_male$stroke_event)
surv_female <- Surv(time = base_female$stroke_time,
                    event = base_female$stroke_event)

## Full model then stepwise model
# Male
library(MASS)

# Male full model
full_male <- coxph(surv_male ~ SYSBP + AGE + DIABETES + BPMEDS,
                  data = base_male)

# Stepwise on male model
step_male <- stepAIC(full_male,
                    scope = list(lower = ~ SYSBP + AGE + DIABETES,
                                upper = ~ SYSBP + AGE + DIABETES + BPMEDS),
                    direction = "both",
                    trace = TRUE)

```

```

## Start:  AIC=661.41
## surv_male ~ SYSBP + AGE + DIABETES + BPMEDS
##
##           Df    AIC
## - BPMEDS   1 660.50
## <none>      661.41
##
## Step:  AIC=660.5
## surv_male ~ SYSBP + AGE + DIABETES
##
##           Df    AIC
## <none>      660.50
## + BPMEDS   1 661.41

```

```
summary(step_male)
```

```

## Call:
## coxph(formula = surv_male ~ SYSBP + AGE + DIABETES, data = base_male)
##
##      n= 1908, number of events= 48
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## SYSBP      0.031160  1.031650 0.005502  5.664 1.48e-08 ***
## AGE        0.058057  1.059776 0.018733  3.099  0.00194 **
## DIABETES  1.297516  3.660195 0.418733  3.099  0.00194 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## SYSBP      1.032      0.9693      1.021      1.043
## AGE        1.060      0.9436      1.022      1.099
## DIABETES   3.660      0.2732      1.611      8.316

```

```
##
## Concordance= 0.789 (se = 0.034 )
## Likelihood ratio test= 63.33 on 3 df, p=1e-13
## Wald test = 79.71 on 3 df, p=<2e-16
## Score (logrank) test = 92.85 on 3 df, p=<2e-16

# female
# Female full model
full_female <- coxph(surv_female ~ SYSBP + AGE + DIABETES + BPMEDS,
                     data = base_female)

# Stepwise on female model
step_female <- stepAIC(full_female,
                      scope = list(lower = ~ SYSBP + AGE + DIABETES,
                                   upper = ~ SYSBP + AGE + DIABETES + BPMEDS),
                      direction = "both",
                      trace = TRUE)

## Start: AIC=838.58
## surv_female ~ SYSBP + AGE + DIABETES + BPMEDS
##
## Df AIC
## <none> 838.58
## - BPMEDS 1 838.59

summary(step_female)

## Call:
## coxph(formula = surv_female ~ SYSBP + AGE + DIABETES + BPMEDS,
## data = base_female)
##
## n= 2434, number of events= 59
##
## coef exp(coef) se(coef) z Pr(>|z|)
## SYSBP 0.02497 1.02528 0.00425 5.874 4.24e-09 ***
## AGE 0.06609 1.06832 0.01918 3.445 0.00057 ***
## DIABETES 0.54124 1.71813 0.52055 1.040 0.29846
## BPMEDS 0.54668 1.72751 0.36746 1.488 0.13683
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## exp(coef) exp(-coef) lower .95 upper .95
## SYSBP 1.025 0.9753 1.0168 1.034
## AGE 1.068 0.9360 1.0289 1.109
## DIABETES 1.718 0.5820 0.6194 4.766
## BPMEDS 1.728 0.5789 0.8407 3.550
##
## Concordance= 0.801 (se = 0.029 )
## Likelihood ratio test= 83.88 on 4 df, p=<2e-16
## Wald test = 92.02 on 4 df, p=<2e-16
## Score (logrank) test = 113.2 on 4 df, p=<2e-16
```

```

## remove bpmeds as not significant in males, barely in females
# add back in cursmoke because we need one extra profile

final_male <- coxph(surv_male ~ SYSBP + AGE +
                    DIABETES + CURSMOKE, data = base_male)
final_female <- coxph(surv_female ~ SYSBP + AGE +
                     DIABETES + CURSMOKE, data = base_female)

### check assumptions

# Schoenfeld residuals
ph_male <- cox.zph(final_male)
ph_female <- cox.zph(final_female)

print(ph_male)

```

```

##          chisq df    p
## SYSBP    2.399  1 0.121
## AGE      0.593  1 0.441
## DIABETES 3.717  1 0.054
## CURSMOKE 0.167  1 0.683
## GLOBAL   5.516  4 0.238

```

```

print(ph_female)

```

```

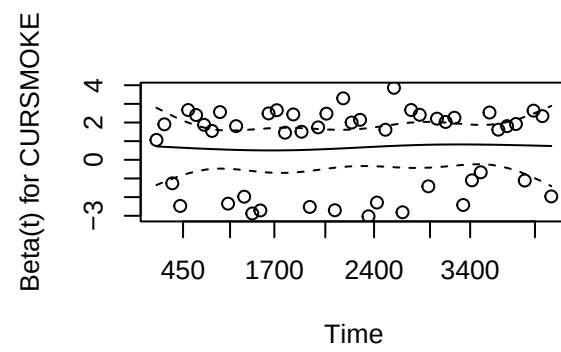
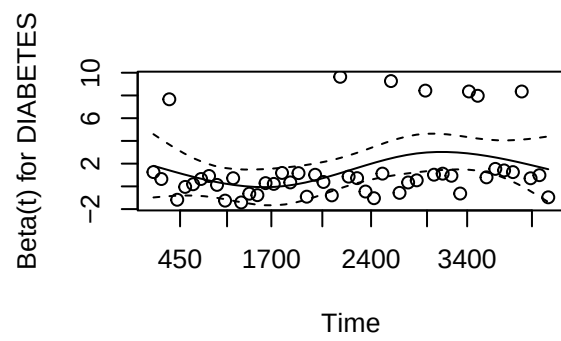
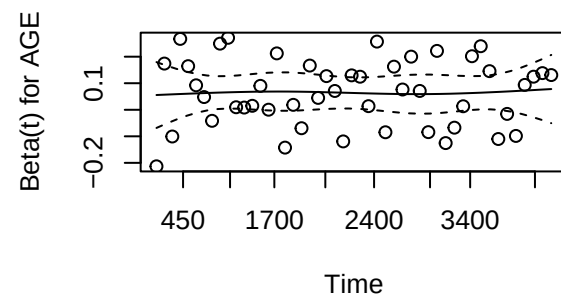
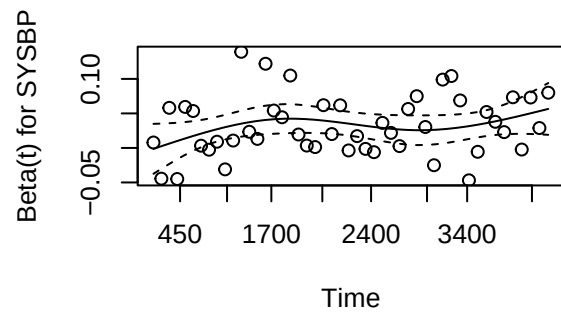
##          chisq df    p
## SYSBP    0.00388  1 0.95
## AGE      0.03515  1 0.85
## DIABETES 0.17667  1 0.67
## CURSMOKE 0.20685  1 0.65
## GLOBAL   0.43919  4 0.98

```

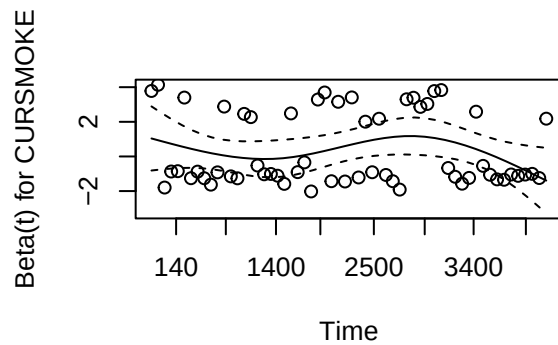
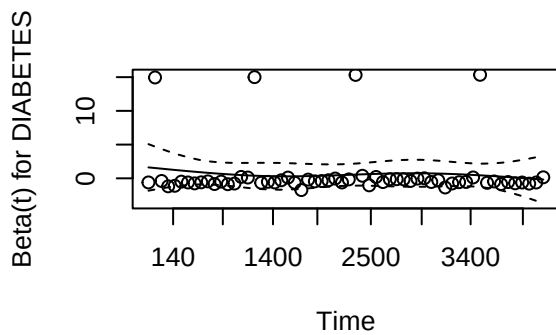
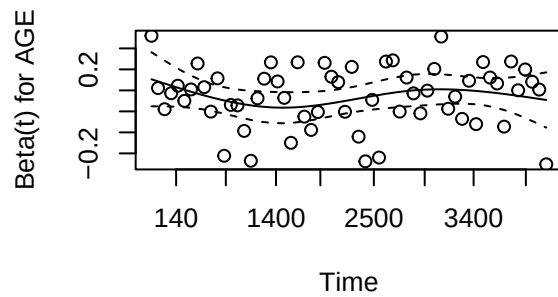
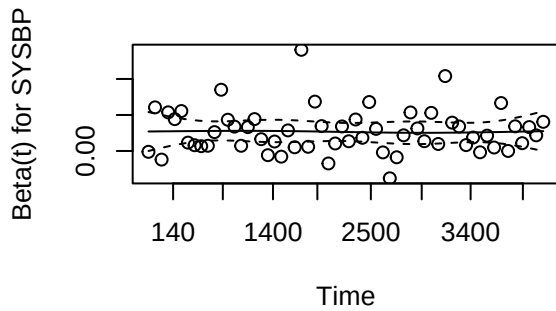
```

# Male
par(mfrow = c(2, 2))
plot(ph_male)

```



```
# Female
par(mfrow = c(2, 2))
plot(ph_female)
```



```
#### Survival Models ####
```

```
## profiles with age
```

```
#40
```

```
profiles_male_40 <- data.frame(
  SYSBP = c(mean(base_male$SYSBP, na.rm = TRUE), 160,
            mean(base_male$SYSBP, na.rm = TRUE), 160,
            mean(base_male$SYSBP, na.rm = TRUE)),
  AGE = 40,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)
```

```
profiles_female_40 <- data.frame(
  SYSBP = c(mean(base_female$SYSBP, na.rm = TRUE), 160,
            mean(base_female$SYSBP, na.rm = TRUE), 160,
            mean(base_female$SYSBP, na.rm = TRUE)),
  AGE = 40,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)
```

```
# Age 40
```

```
male_40 <- as.vector(summary(survfit(final_male, newdata = profiles_male_40),
                             times = 3652.5)$surv)
```

```

female_40 <- as.vector(summary(survfit(final_female, newdata = profiles_female_40),
                                times = 3652.5)$surv)

table_40 <- data.frame(
  Age      = 40,
  Profile  = c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"),
  Male     = round(1 - male_40, 4),
  Female   = round(1 - female_40, 4)
)

# 50
profiles_male_50 <- data.frame(
  SYSBP    = c(mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE)),
  AGE      = 50,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)

profiles_female_50 <- data.frame(
  SYSBP    = c(mean(base_female$SYSBP, na.rm = TRUE), 160,
               mean(base_female$SYSBP, na.rm = TRUE), 160,
               mean(base_female$SYSBP, na.rm = TRUE)),
  AGE      = 50,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)

# Age 50
male_50 <- as.vector(summary(survfit(final_male, newdata = profiles_male_50),
                             times = 3652.5)$surv)
female_50 <- as.vector(summary(survfit(final_female, newdata = profiles_female_50),
                                times = 3652.5)$surv)

table_50 <- data.frame(
  Age      = 50,
  Profile  = c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"),
  Male     = round(1 - male_50, 4),
  Female   = round(1 - female_50, 4)
)

# 60
profiles_male_60 <- data.frame(
  SYSBP    = c(mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE), 160,
               mean(base_male$SYSBP, na.rm = TRUE)),
  AGE      = 60,
  DIABETES = c(0, 0, 1, 1, 0),
  CURSMOKE = c(0, 0, 0, 0, 1)
)

```

```

)

profiles_female_60 <- data.frame(
  SYSBP      = c(mean(base_female$SYSBP, na.rm = TRUE), 160,
                  mean(base_female$SYSBP, na.rm = TRUE), 160,
                  mean(base_female$SYSBP, na.rm = TRUE)),
  AGE        = 60,
  DIABETES   = c(0, 0, 1, 1, 0),
  CURSMOKE   = c(0, 0, 0, 0, 1)
)

# Age 60
male_60      <- as.vector(summary(survfit(final_male, newdata = profiles_male_60),
                                   times = 3652.5)$surv)
female_60    <- as.vector(summary(survfit(final_female, newdata = profiles_female_60),
                                   times = 3652.5)$surv)

table_60 <- data.frame(
  Age        = 60,
  Profile     = c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"),
  Male        = round(1 - male_60, 4),
  Female      = round(1 - female_60, 4)
)

##### DO NOT USE THIS TABLE #####
# Final table for all of them

final_table <- rbind(table_40, table_50, table_60)
rownames(final_table) <- NULL

print(final_table)

```

```

##   Age      Profile   Male Female
## 1   40      Average 0.0056 0.0060
## 2   40      High BP 0.0139 0.0121
## 3   40      Diabetes 0.0247 0.0107
## 4   40 High BP + DM 0.0602 0.0216
## 5   40      Smoker 0.0110 0.0084
## 6   50      Average 0.0107 0.0122
## 7   50      High BP 0.0264 0.0245
## 8   50      Diabetes 0.0466 0.0217
## 9   50 High BP + DM 0.1117 0.0436
## 10  50      Smoker 0.0208 0.0170
## 11  60      Average 0.0203 0.0247
## 12  60      High BP 0.0497 0.0494
## 13  60      Diabetes 0.0869 0.0439
## 14  60 High BP + DM 0.2022 0.0869
## 15  60      Smoker 0.0394 0.0344

```

```
#####
```

```

# Try a different approach
# Age 40
m40 <- summary(survfit(final_male, newdata = profiles_male_40),
               times = 3652.5)$surv
f40 <- summary(survfit(final_female, newdata = profiles_female_40),
               times = 3652.5)$surv

# Age 50
m50 <- summary(survfit(final_male, newdata = profiles_male_50),
               times = 3652.5)$surv
f50 <- summary(survfit(final_female, newdata = profiles_female_50),
               times = 3652.5)$surv

# Age 60
m60 <- summary(survfit(final_male, newdata = profiles_male_60),
               times = 3652.5)$surv
f60 <- summary(survfit(final_female, newdata = profiles_female_60),
               times = 3652.5)$surv

# Build table manually indexing each profile [1,i]
final_table <- data.frame(
  Age      = rep(c(40, 50, 60), each = 5),
  Profile   = rep(c("Average", "High BP", "Diabetes", "High BP + DM", "Smoker"), 3),
  Male      = round(1 - c(m40[1,1], m40[1,2], m40[1,3], m40[1,4], m40[1,5],
                          m50[1,1], m50[1,2], m50[1,3], m50[1,4], m50[1,5],
                          m60[1,1], m60[1,2], m60[1,3], m60[1,4], m60[1,5]), 4),
  Female    = round(1 - c(f40[1,1], f40[1,2], f40[1,3], f40[1,4], f40[1,5],
                          f50[1,1], f50[1,2], f50[1,3], f50[1,4], f50[1,5],
                          f60[1,1], f60[1,2], f60[1,3], f60[1,4], f60[1,5]), 4)
)

print(final_table)

```

```

##      Age      Profile  Male Female
## 1    40      Average 0.0056 0.0060
## 2    40      High BP 0.0139 0.0121
## 3    40      Diabetes 0.0247 0.0107
## 4    40 High BP + DM 0.0602 0.0216
## 5    40      Smoker 0.0110 0.0084
## 6    50      Average 0.0107 0.0122
## 7    50      High BP 0.0264 0.0245
## 8    50      Diabetes 0.0466 0.0217
## 9    50 High BP + DM 0.1117 0.0436
## 10   50      Smoker 0.0208 0.0170
## 11   60      Average 0.0203 0.0247
## 12   60      High BP 0.0497 0.0494
## 13   60      Diabetes 0.0869 0.0439
## 14   60 High BP + DM 0.2022 0.0869
## 15   60      Smoker 0.0394 0.0344

```

```

# Add a Sex label column and blank separator rows for grouping
surv_table_display <- data.frame(
  Profile = c(

```

```

    "Male",
    "  Average", "  High BP (160)", "  Diabetes", "  High BP + DM", "  Smoker",
    "Female",
    "  Average", "  High BP (160)", "  Diabetes", "  High BP + DM", "  Smoker"
  ),
  Age_40 = c(
    "",
    round(m40[1,1], 4), round(m40[1,2], 4), round(m40[1,3], 4), round(m40[1,4], 4), round(m40[1,5], 4),
    "",
    round(f40[1,1], 4), round(f40[1,2], 4), round(f40[1,3], 4), round(f40[1,4], 4), round(f40[1,5], 4)
  ),
  Age_50 = c(
    "",
    round(m50[1,1], 4), round(m50[1,2], 4), round(m50[1,3], 4), round(m50[1,4], 4), round(m50[1,5], 4),
    "",
    round(f50[1,1], 4), round(f50[1,2], 4), round(f50[1,3], 4), round(f50[1,4], 4), round(f50[1,5], 4)
  ),
  Age_60 = c(
    "",
    round(m60[1,1], 4), round(m60[1,2], 4), round(m60[1,3], 4), round(m60[1,4], 4), round(m60[1,5], 4),
    "",
    round(f60[1,1], 4), round(f60[1,2], 4), round(f60[1,3], 4), round(f60[1,4], 4), round(f60[1,5], 4)
  )
)

cat("\\newpage")

```

```
## \newpage
```

```

kable(surv_table_display,
      col.names = c("Risk Profile", "Age 40", "Age 50", "Age 60"),
      caption   = "10-Year Stroke-Free Survival Probability by Risk Profile, Age, and Sex",
      booktabs  = TRUE,
      align     = c("l", "c", "c", "c"))

```

Table 2: 10-Year Stroke-Free Survival Probability by Risk Profile, Age, and Sex

Risk Profile	Age 40	Age 50	Age 60
Male			
Average	0.9944	0.9893	0.9797
High BP (160)	0.9861	0.9736	0.9503
Diabetes	0.9753	0.9534	0.9131
High BP + DM	0.9398	0.8883	0.7978
Smoker	0.989	0.9792	0.9606
Female			
Average	0.994	0.9878	0.9753
High BP (160)	0.9879	0.9755	0.9506
Diabetes	0.9893	0.9783	0.9561
High BP + DM	0.9784	0.9564	0.9131
Smoker	0.9916	0.983	0.9656

```
#####
# Secondary Analysis
#####

# Prepare periods data from raw fhsdata
# Factor the period and categorical variables - base R only
periods_data <- fhsdata

periods_data$PERIOD <- factor(fhsdata$PERIOD, levels = c(1, 2, 3),
                             labels = c("Period 1", "Period 2", "Period 3"))
periods_data$DIABETES <- factor(fhsdata$DIABETES, levels = c(0, 1), labels = c("No", "Yes"))
periods_data$CURSMOKE <- factor(fhsdata$CURSMOKE, levels = c(0, 1), labels = c("No", "Yes"))

# Build the table stratified by period
periods_table <- tbl_summary(
  data = periods_data[, c("PERIOD", "AGE", "SYSBP", "DIABETES", "CURSMOKE")],
  by = PERIOD,
  label = list(
    AGE ~ "Age (years)",
    SYSBP ~ "Systolic Blood Pressure (mmHg)",
    DIABETES ~ "Diabetes",
    CURSMOKE ~ "Current Smoker"
  ),
  statistic = list(
    all_continuous() ~ "{mean} ({sd})",
    all_categorical() ~ "{n} ({p}%)"
  ),
  digits = list(
    all_continuous() ~ 1,
    all_categorical() ~ c(0, 1)
  ),
  missing = "ifany",
  missing_text = "Missing"
) %>%
add_overall() %>%
add_p(
  test = list(
    all_continuous() ~ "oneway.test",
    all_categorical() ~ "chisq.test"
  )
) %>%
bold_labels() %>%
bold_p(t = 0.05) %>%
modify_header(label ~ "**Characteristic**") %>%
modify_caption("**Table 3. Risk Factor Comparison Across Data Collection Periods**")

cat("\n\\newpage")

```

```
## \newpage
```

```
kable(as_tibble(periods_table),
      caption = "Risk Factor Comparison Across Data Collection Periods",
      booktabs = TRUE,
      align    = "l",
      table.envir = "table[H]")
```

Table: Risk Factor Comparison Across Data Collection Periods

Characteristic	Overall	Period 1	Period 2	Period 3	p-value
N = 11,627					
N = 4,434					
N = 3,930					
N = 3,263					
Age (years)	54.8 (9.6)	49.9 (8.7)	55.4 (8.5)	60.6 (8.3)	<0.001
Systolic Blood Pressure (mmHg)	136.3 (22.8)	132.9 (22.4)	136.9 (22.5)	140.2 (22.9)	<0.001
Diabetes	530 (4.6%)	121 (2.7%)	155 (3.9%)	254 (7.8%)	<0.001
Current Smoker	5,029 (43.3%)	2,181 (49.2%)	1,727 (43.9%)	1,121 (34.4%)	<0.001